

SRI KRISHNA KALYAN BANDARU

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ABOUT ME

Master of Artificial Intelligence and Machine Learning graduate from the University of Adelaide with hands-on experience applying data analytics and AI techniques to solve real-world problems. End-to-end data and AI solutions delivered — from dataset construction and exploratory analysis through to model development, production deployment, and stakeholder insight communication. Proven ability to explore complex datasets, apply AI techniques, and translate findings into clear, actionable advice for technical and non-technical audiences. Collaborative, curious, and driven by impact — eager to join PwC's Data Analytics & AI team in Sydney to help clients harness data for competitive advantage and confident decision-making.

EDUCATION

Master of Artificial Intelligence & Machine Learning Dec 2025
University of Adelaide | **GPA: 6.1** | Focus: Computer Vision, Deep Learning, NLP, MLOps

Master of Science in Mechatronics Engineering 2022
University of Rome 'Tor Vergata' | **Grade: 106/110** | Focus: Control Systems, MATLAB/Simulink

Bachelor of Technology in Mechanical Engineering 2018
KL University, India | **CGPA: 8.89/10, First Class Distinction** | Focus: Engineering Fundamentals, C Programming, Data Structures

TECHNICAL SKILLS

Languages	Python, SQL, C/C++, Java, MATLAB/Simulink, Git
AI & ML	PyTorch, TensorFlow, Scikit-learn, YOLO, Mask-RCNN, BERT, Transfer Learning, NLP
Data & Analytics	Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Statistical Analysis, Exploratory Data Analysis
Deployment & Cloud	Docker, FastAPI, CI/CD Pipelines, Azure, Linux, REST APIs, Streamlit
Systems & IoT	ROS2, Gazebo, Jetson Nano, STM32, Edge AI, Embedded Systems, Autonomous Navigation

EXPERIENCE

Research Assistant — Auto-ID Lab, University of Adelaide Feb 2025 – Aug 2025
Adelaide, Australia — Autonomous Systems & Computer Vision

- Faced with an outdated ROS1 UAV codebase incompatible with modern hardware, led full migration to ROS2 — restoring system compatibility and enabling deployment on current drone platforms
- To validate AI algorithms safely before real-world flight, built a complete Gazebo indoor simulation environment — reducing hardware risk and accelerating iteration cycles for the research team
- Tasked with deploying AI workloads to resource-constrained edge devices, containerised and deployed full inference stack onto Jetson Nano/Orin — achieving stable, production-ready autonomous drone operations
- Evaluated YOLO architectures (YOLOv5s, YOLOv12s) against VisDrone aerial datasets, achieving **mAP50 of 0.642** for real-time person and vehicle detection in complex aerial scenes

Machine Learning Engineer Intern — Proto Corp Pvt Ltd Aug 2024 – Nov 2024
Remote, India — Computer Vision & MLOps

- To address a lack of labelled training data, collected, cleaned, and annotated **3,500+ construction defect images** at pixel-level precision across 5 categories — enabling the first production-ready CV pipeline for the client
- Evaluated four model architectures (YOLOv8m-seg, YOLOv9c-seg, YOLOv11m-seg, Mask R-CNN) through rigorous comparative testing — selected YOLOv8m-seg for production based on accuracy and inference speed
- Built a FastAPI inference backend and Streamlit dashboard enabling non-technical stakeholders to run real-time segmentation — removing bottlenecks in the client's quality inspection workflow

Project Engineer — Wipro Technologies May 2018 – Sept 2019
Hyderabad, India — IT Operations & SAP PM Support

- Managed plant maintenance operations across UK industrial clients using SAP PM — led a team of four to consistently meet SLA targets and minimise equipment downtime
- Identified recurring escalation patterns through root-cause documentation, reducing repeated incidents and improving team knowledge base quality

KEY PROJECTS

Autonomous UAV Object Detection System — ROS2, YOLOv12s, Jetson Nano, EGO-Planner, Gazebo
Developed modular ROS2 architecture combining EGO-Planner obstacle avoidance with YOLO real-time object detection for disaster response applications, validated via ArduPilot SITL hardware-in-the-loop simulation, achieving **mAP50 of 0.642** on VisDrone and **~2,000 features/frame** with SuperPoint — deployed end-to-end on live drone hardware.

Embedded Neural Network on STM32 (On-Device AI) — TensorFlow Lite, STM32CubeMX, X-CUBE-AI, C/C++
Applied model compression and weight quantisation (~3× reduction) to deploy a CNN directly onto STM32H747I-DISCO micro-controller, eliminating cloud dependency for low-connectivity environments and achieving **97% accuracy** across 18 categories at **11.2 FPS** fully on-device.

Alzheimer's Disease Diagnosis from MEG Signals — PyTorch, AlexNet, ResNet, GoogLeNet, SVM
Engineered MEG time-series neurological data into 2D representations and applied transfer learning across multiple architectures with leave-one-patient-out cross-validation, achieving **81% accuracy** and **AUC of 0.74** for early-stage Alzheimer's diagnosis.